Proxy Features and Fairness

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Responsible Plant Classifier

WORKBOOK · 1 OF 21

Recall and choose your starting point

> Record your starting explanation for Proxy Features and Fairness before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Subgroup Performance

Without opening the old lesson, explain one key idea from Subgroup Performance. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Feature Engineering

Without opening the old lesson, explain one key idea from Feature Engineering. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Reproducible Experiments

Without opening the old lesson, explain one key idea from Reproducible Experiments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

WORKBOOK · 2 OF 21

Practise with fading help

> Complete the supported Proxy Features and Fairness trace and record the first step you can explain without help.

Use W2: Find a proxy risk for **Proxy feature**s and **fairness**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

WORKBOOK · 3 OF 21

Individual reasoning check

> Check the proposed Proxy Features and Fairness explanation against the supplied facts before discussing it.

Use the worked case for **Proxy feature**s and **fairness**. Proposed explanation: “A final answer is enough”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 21

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply independently

A team removes postcode but retains school code and claims all location-related information is gone. Evaluate the claim and request **evidence**.

> **My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 21

Your learning target

> Connect the Proxy Features and Fairness target to a trace and a justified conclusion.

A leaf model performs well using location ID alone. Explain a possible proxy problem and design a diagnostic evaluation.

WORKBOOK · 6 OF 21

Solve the given case

> Solve the supplied Proxy Features and Fairness case and show the deciding evidence.

Background colour tracks collection location, associated with plant types.

Explain proxy risk and test.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

WORKBOOK · 7 OF 21

Check a changed case

Apply the procedure to the changed case and record why the conclusion follows.

A plant classifier uses camera_model, and one species was photographed mostly on a single camera. What shortcut might it learn and how could you investigate?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 32 Test and reflect

WORKBOOK · 8 OF 21

Design a revealing test

> Specify a changed input, expected result and observation separately.

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

Project checkpoint

> Justify what Proxy Features and Fairness adds to Responsible Plant Classifier and keep the evidence.

Document Plant Classifier proxy risks, paired-test **evidence**, **mitigation**, and remaining **fairness** limitation.

The project change and **evidence** I will keep

Write or draw your reasoning and evidence.

WORKBOOK · 10 OF 21

Self check

> Judge explanation, evidence, transfer and limits separately; attach the work behind your judgement.

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

WORKBOOK · 11 OF 21

Transfer practice from the original programme

> Use this earlier case as transfer practice. A case already discussed with feedback is not fresh assessment evidence.

A leaf model performs well using location ID alone. Explain a possible proxy problem and design a diagnostic evaluation.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

WORKBOOK · 12 OF 21

W1 Read an association

Read an association. Apply the supplied Proxy Features and Fairness facts and keep your reasoning.

What do the device counts support, and what do they not support?

My reasoning and evidence

WORKBOOK · 13 OF 21

W2 Find a proxy risk

> Find a proxy risk. Apply the supplied Proxy Features and Fairness facts and keep your reasoning.

> Why might removing building but keeping device code leave related information?

My reasoning and evidence

W3 Plan an ablation

> Plan an ablation. Apply the supplied Proxy Features and Fairness facts and keep your reasoning.

Describe a controlled with/without-code comparison.

My reasoning and evidence

WORKBOOK · 15 OF 21

W4 Apply independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply independently. Apply the supplied Proxy Features and Fairness facts and keep your reasoning.

A team removes postcode but retains school code and claims all location-related information is gone. Evaluate the claim and request **evidence**.

My reasoning and evidence

WORKBOOK · 16 OF 21

W5 Limit the result

> Limit the result. Apply the supplied Proxy Features and Fairness facts and keep your reasoning.

Does unchanged accuracy after removing a proxy prove **fairness**?

My reasoning and evidence

WORKBOOK · 17 OF 21

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 18 OF 21

Pictorial case practice: Proxy Features and Fairness

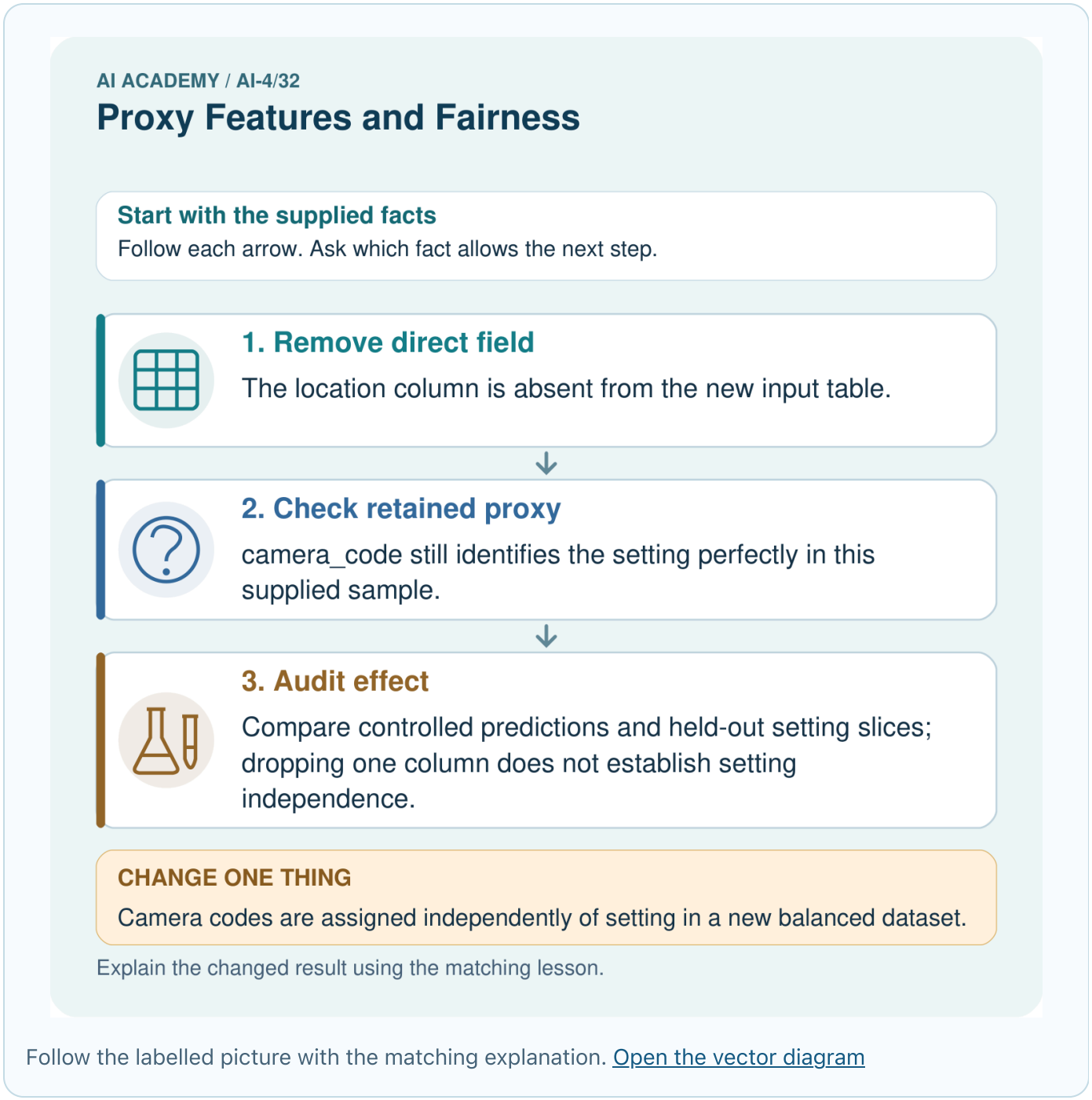
> Removing an obvious setting field may leave another field that carries the same information.

> Explain the labelled step before checking the changed condition.

Removing an obvious setting field may leave another field that carries the same information.

Toy table has location=greenhouse/desk and camera_code=G/D. Every G record is greenhouse; every D record is desk. The team drops location.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

What **evidence** contradicts the original “all setting information removed” claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Responsible Plant Classifier record.

Project: Responsible Plant Classifier

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	

Evidence to keep	My record
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

WORKBOOK · 20 OF 21

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Removing an obvious setting field may leave another field that carries the same information.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

1. Remove direct field

2. Audit effect

3. Check retained proxy

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Audit effect Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence. Read the picture in order.

Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence. The arrows show which recorded **input** or intermediate result each decision uses.

Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Camera codes are assigned independently of setting in a new balanced dataset. Which conclusion is supported by the supplied facts?

1. camera_code no longer perfectly identifies setting in that dataset.

2. We can decide the result without examining the changed **input** or the procedure.

3. This one example guarantees correct results for every future **input**.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: camera_code no longer perfectly identifies setting in that dataset. The observed mapping changed; other retained fields may still contain setting information.

Repair a missing building block

Review Subgroup Performance (ai-4/31). Goal: Calculate subgroup results with sample sizes. Short practice: Why should the report retain subgroup rates alongside the total?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Feature Engineering (ai-4/29). Goal: Create features available at **prediction** time. Short practice: What should be saved so another group transforms the query identically?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Reproducible Experiments (ai-4/26). Goal: Record enough detail to reproduce a run. Short practice: What should accompany the mean **70%**?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

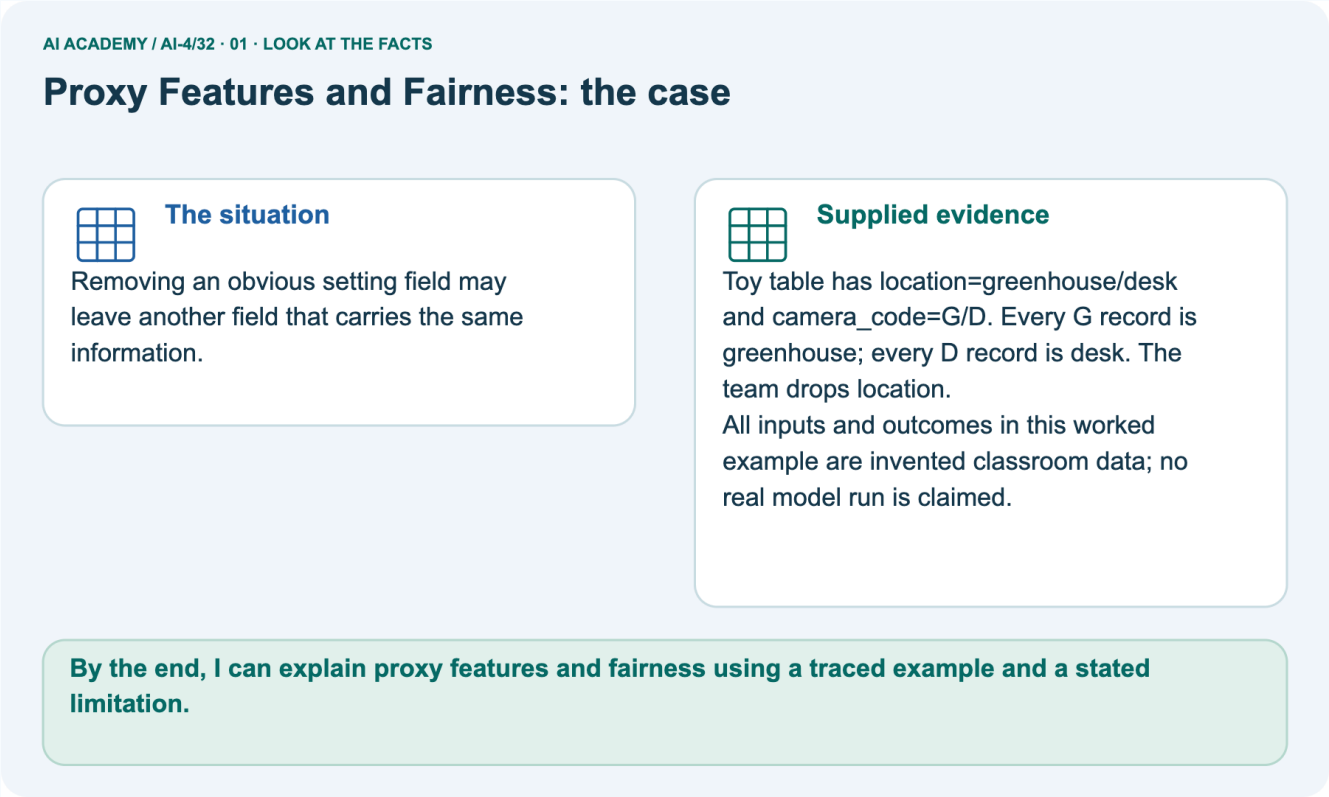
Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

Visual concept journey

> Explain each Proxy Features and Fairness picture from its labels, then record a relevant limit.

01. Proxy Features and Fairness: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

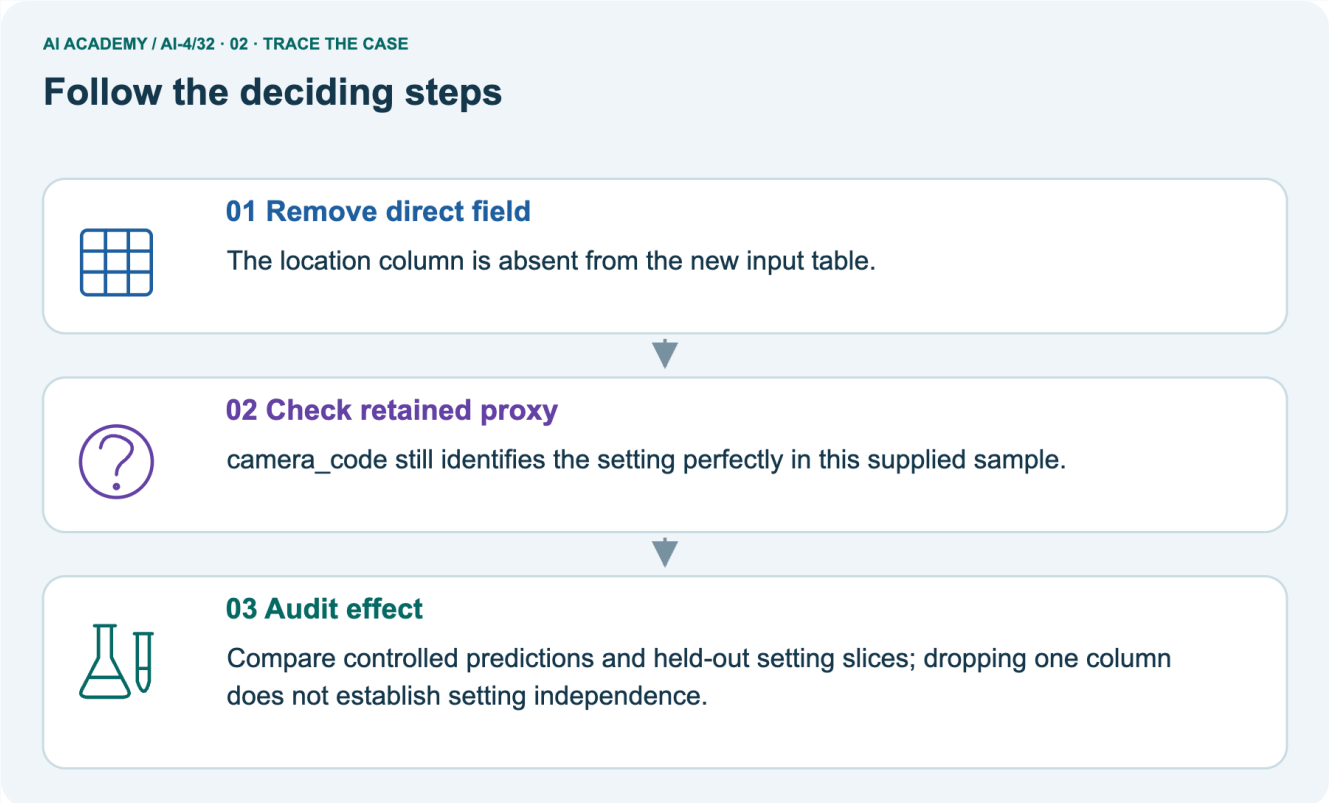
Removing an obvious setting field may leave another field that carries the same information. Supplied facts: Toy table has location=greenhouse/desk and camera_code=G/D. Every G record is greenhouse; every D record is desk.

The team drops location. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



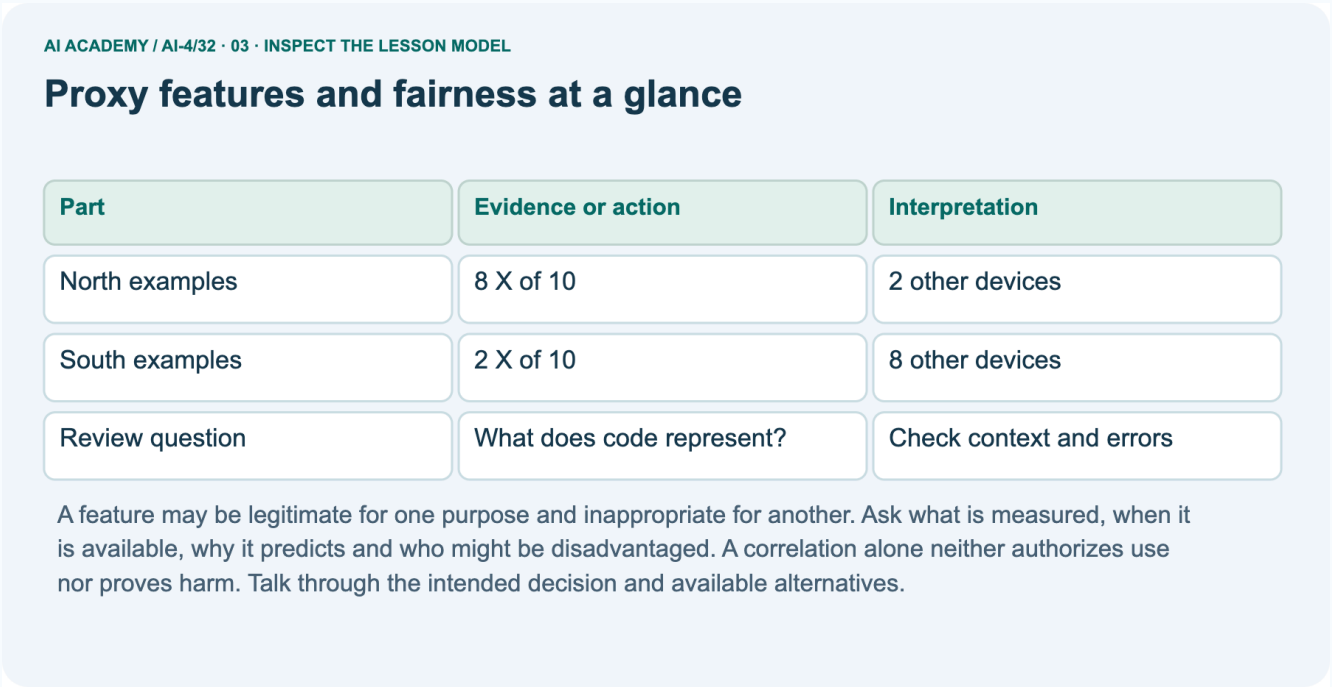
Follow the labelled picture with the matching explanation. Open the vector diagram

Remove direct field: The location column is absent from the new **input** table. → Check retained proxy: camera_code still identifies the setting perfectly in this supplied sample. → Audit effect: Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Proxy features and fairness at a glance



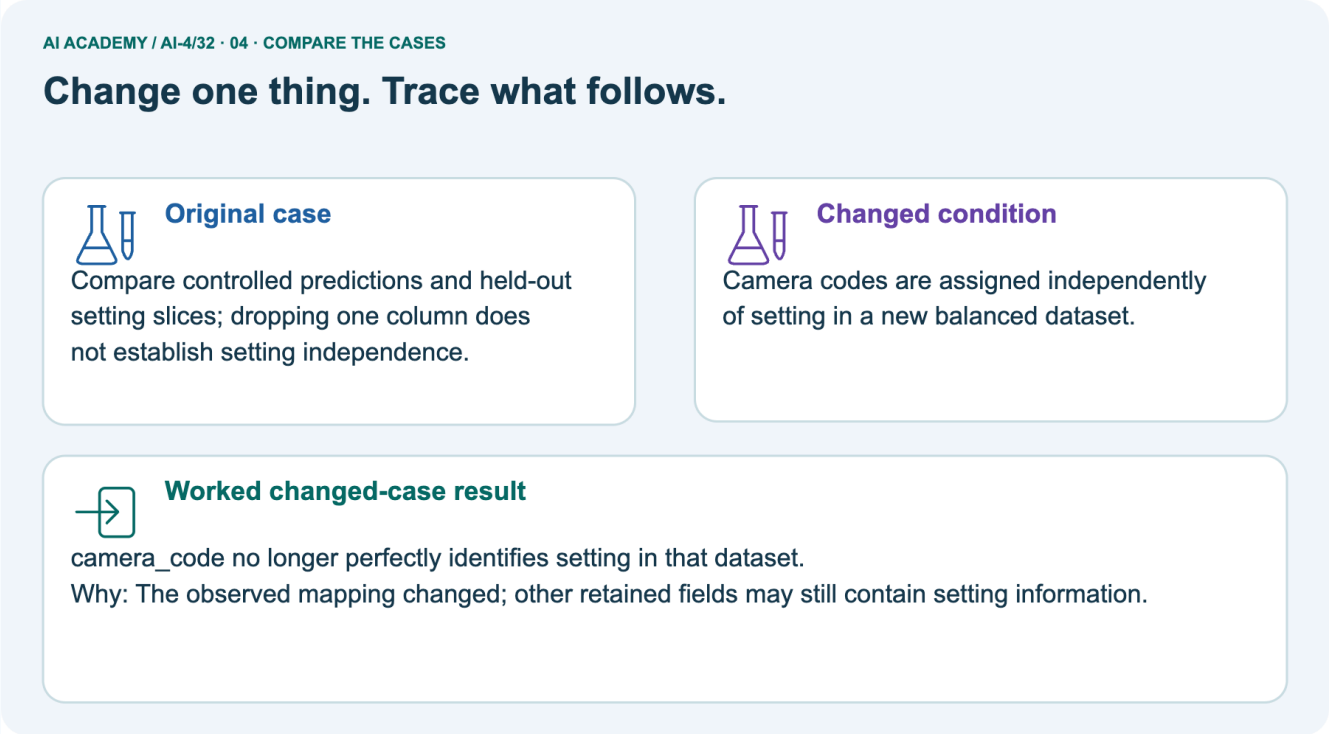
Follow the labelled picture with the matching explanation. Open the vector diagram

Proxy features and **fairness** at a glance. North examples | 8 X of 10 | 2 other devices. South examples | 2 X of 10 | 8 other devices. Review question | What does code represent? | Check context and errors

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

Changed condition: Camera codes are assigned independently of setting in a new balanced dataset. Expected result: camera_code no longer perfectly identifies setting in that dataset.

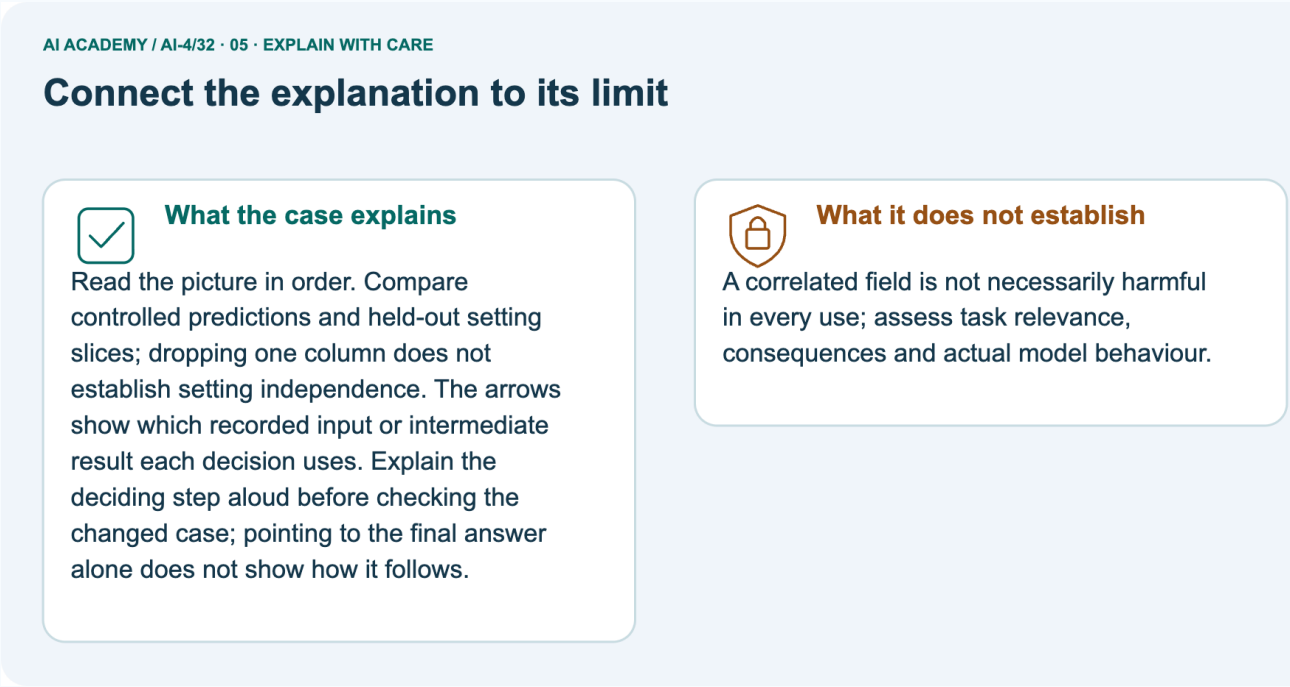
Reason: The observed mapping changed; other retained fields may still contain setting information.

Predict before checking the worked changed-case result. Name what changed and what the

supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. Compare controlled **prediction**s and held-out setting slices; dropping one column does not establish setting independence.

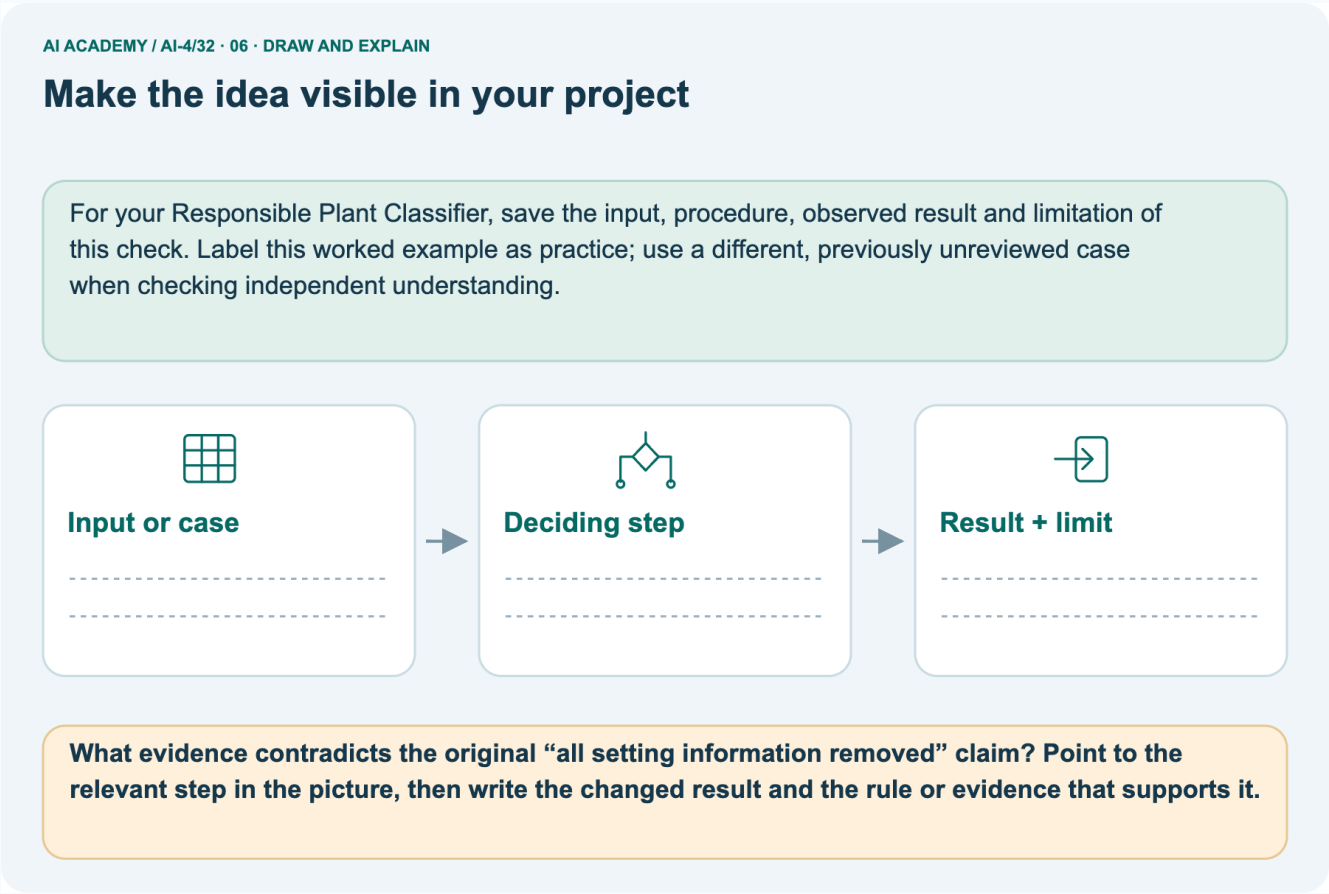
The arrows show which recorded **input** or intermediate result each decision uses. Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A correlated field is not necessarily harmful in every use; assess task relevance, consequences and actual model behaviour.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Responsible Plant Classifier, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What **evidence** contradicts the original “all setting information removed” claim? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.